PPP tube power amplifier, 4x KT88 parallel push–pull (circlotron), mono block

Amplifier

Datei: amp.kicad_sch

SMPS primary

Datei: smps_pri.kicad_sch

SMPS outputs

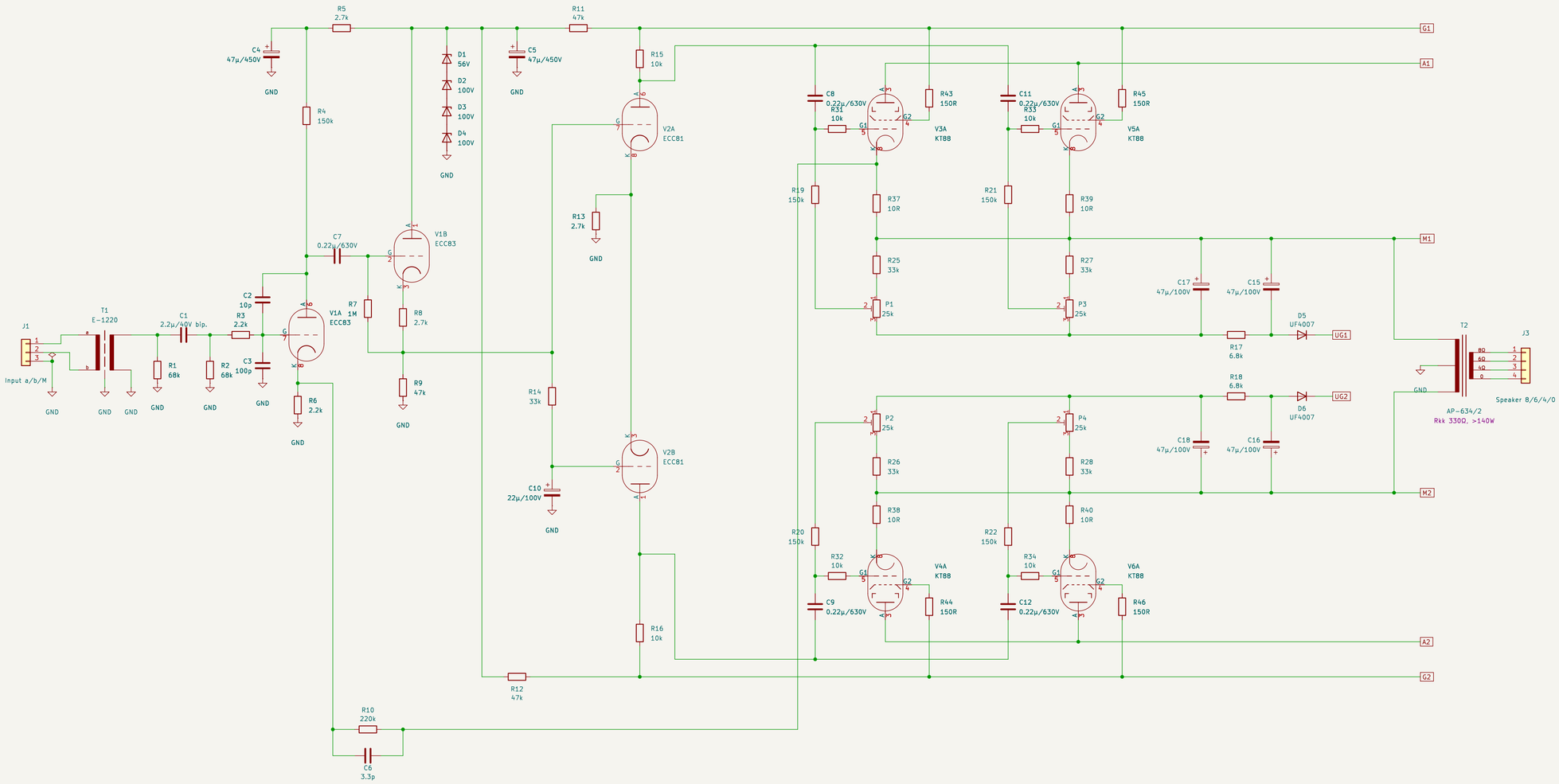
Datei: smps_sec.kicad_sch

Sheet: /
File: ppp.kicad_sch

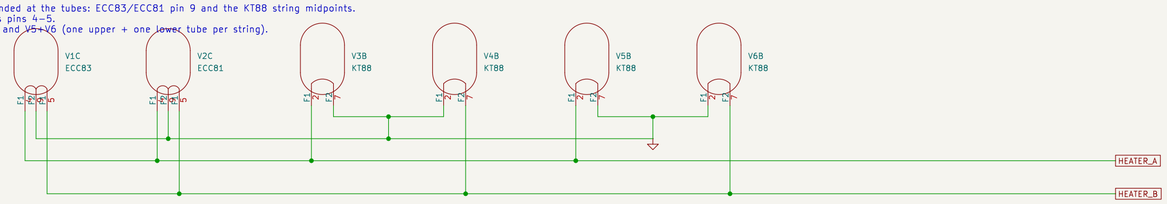
Title: PPP power amplifier 4x KT88 (mono)

Size: A4	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. 9.0.8		Id: 1/4

PPP 4x KT88 – Amplifier



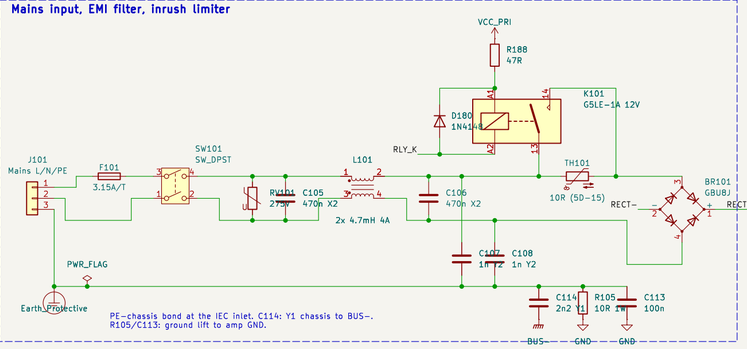
12.6V- heaters, centre grounded at the tubes: ECC83/ECC81 pin 9 and the KT88 string midpoints.
ECC83/ECC81: 12.6V across pins 4–5.
KT88 in series pairs V3+V4 and V5+V6 (one upper + one lower tube per string).



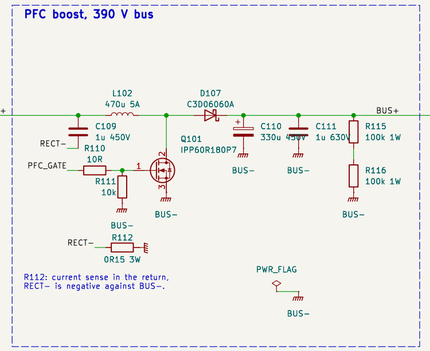
Original designators: V1=R81 (ECC83), V2=R82 (ECC81), V3..V6=R83..R86 (KT88 variant, original EL34: R87/R88 omitted). T1=Ü1 (E-1220), T2=Ü2 (AP-634/2).
Supply rails A1/G1/M1, A2/G2/M2 and UG1/UG2 come from the power supply sheet.
Upper tubes: anode A1, screen G1, cathode M1 (G1 is referenced to M1).
Lower tubes: anode A2, screen G2, cathode M2 (G2 is referenced to M2).
Ug1/Ug2: bias windings, floating on M1/M2: bias adjust P1..P4.

Sheet: /Amplifier/	
File: amp.kicad_sch	
Title: PPP power amplifier 4x KT88 – amplifier (mono)	
Size: A2	Date: 2026-10-20
KiCad E.D.A. 9.0.8	Id: 2/4
	Rev: 1

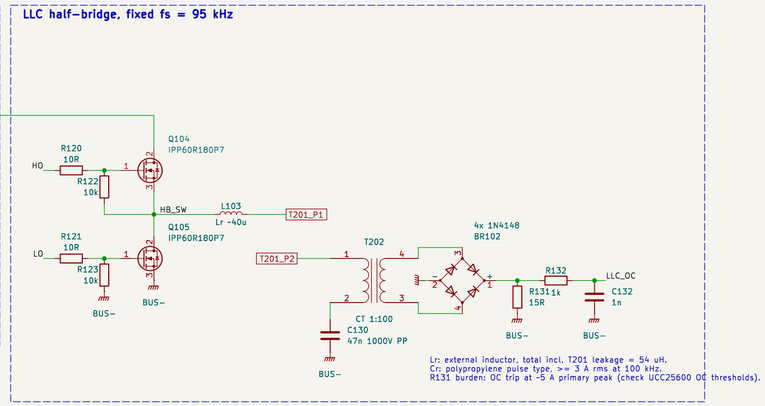
Mains input, EMI filter, inrush limiter



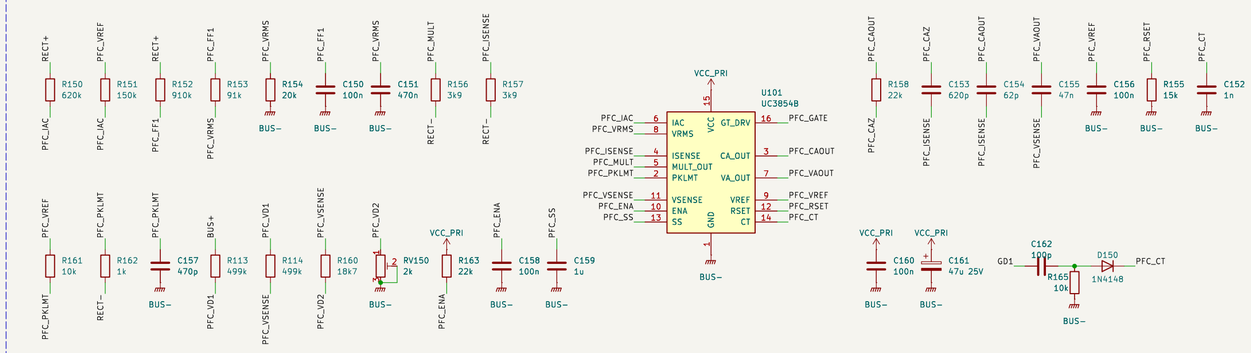
PFC boost, 390 V bus



LLC half-bridge, fixed fs = 95 kHz

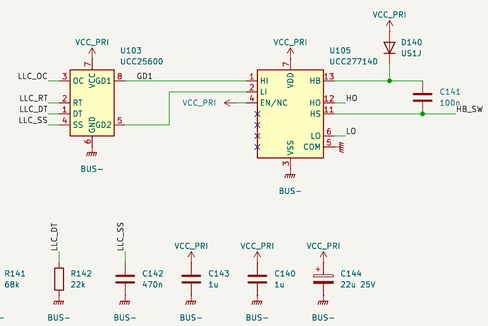


PFC control (UC3854B, average current mode, start values from TI app note U-134)



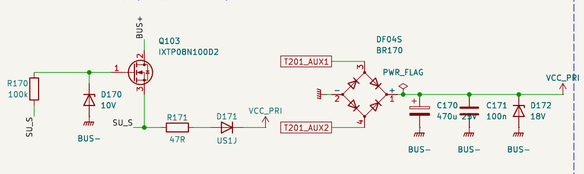
CT 1n / RSET 15k: free-running ~83 kHz, synced to the LLC clock (95 kHz) via C162/D150 from U103 GD1.
VSENSE 7.5 V: Vbus = 7.5 V x (R113+R114+R160+RV150)/(R160+RV150). RV150 sets 370…408 V. R113/R114/R150/R152: HV types (0.5 W, >= 500 V).
Voltage loop: add Rvf/Cvf per U-134 once the loop is calculated. Pinout of the UC3854B symbol: verify against the data sheet.

LLC control (UCC25600, open loop = fixed frequency) and gate driver



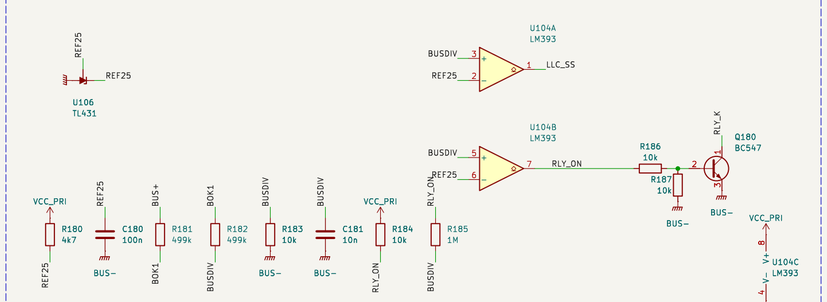
R141 sets fs (~95 kHz, slightly below fr ~100 kHz), R142 the dead time (~200 ns):
check the UCC25600 data sheet equations. SS sweeps from high frequency down to fs.
U104A holds LLC_SS low until Vbus > 250 V. GD1 also syncs the PFC oscillator.

Startup and aux supply (VCC_PRI ~14 V)



Q103 (depletion mode) starts the controllers from the bus;
it turns off when W2 (aux) lifts VCC_PRI above ~12 V.

Bus OK (> 250 V): NTC bypass relay, LLC enable



U104A: open collector holds LLC_SS low while the bus is low. U104B + Q180: relay K101 bypasses TH101
once the bus is charged. R185: hysteresis. R181/R182: HV types.

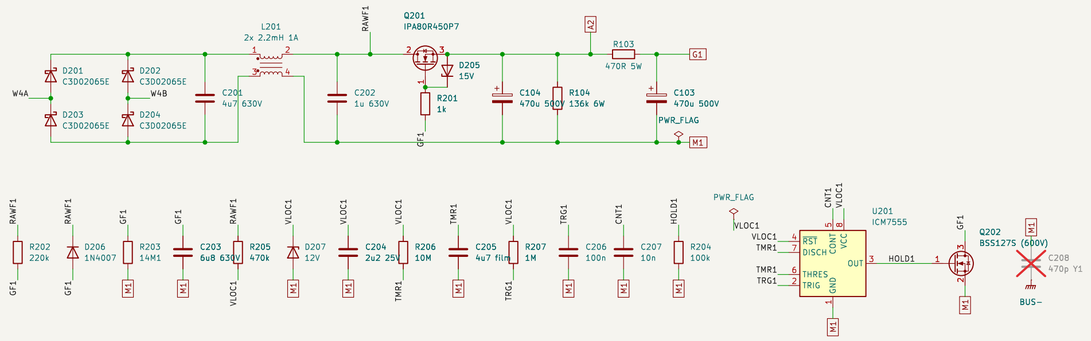
Primary side is at mains potential
Draft — values are start values, see SMPS.md

Sheet: /SMPS primary/
File: smps_pri.kicad_sch

Title: PPP power amplifier 4x KT88 — SMPS primary (mono)

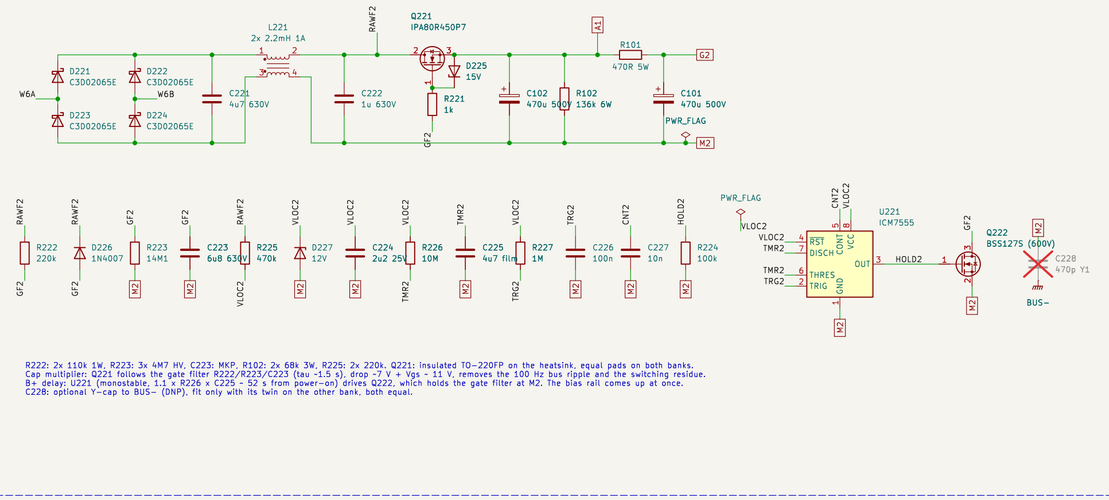
Size: A2	Date: 2026-10-20	Rev: 1
KiCad E.D.A. 9.0.8		Id: 3/4

Bank M1: W4 -> A2 / G1, referenced to M1



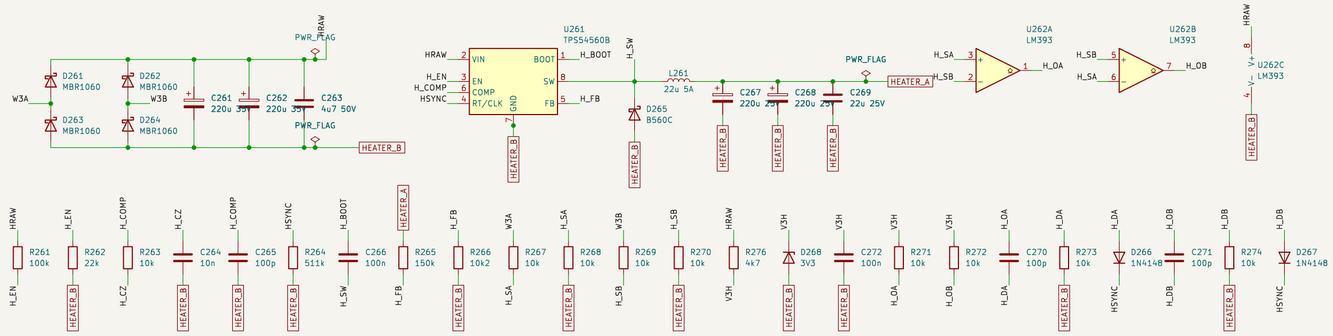
R202: 2x 110k 1W, R203: 3x 4M7 HV, C203: MKP, R104: 2x 68k 3W, R205: 2x 220k. Q201: insulated TO-220FP on the heatsink, equal pads on both banks.
Cap multiplier: Q201 follows the gate filter R202/R203/C203 (tau ~1.5 s), drop ~7 V + Vgs - 11 V, removes the 100 Hz bus ripple and the switching residue.
B+ delay: U201 (monostable, 1.1 x R206 x C205 ~ 52 s from power-on) drives Q202, which holds the gate filter at M1. The bias rail comes up at once.
C208: optional Y-cap to BUS- (DNP), fit only with its twin on the other bank, both equal.

Bank M2: W6 -> A1 / G2, referenced to M2



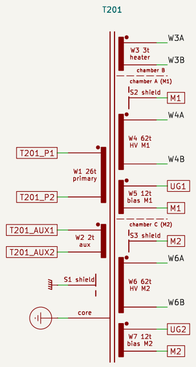
R222: 2x 110k 1W, R223: 3x 4M7 HV, C223: MKP, R102: 2x 68k 3W, R225: 2x 220k. Q221: insulated TO-220FP on the heatsink, equal pads on both banks.
Cap multiplier: Q221 follows the gate filter R222/R223/C223 (tau ~1.5 s), drop ~7 V + Vgs - 11 V, removes the 100 Hz bus ripple and the switching residue.
B+ delay: U221 (monostable, 1.1 x R226 x C225 ~ 52 s from power-on) drives Q222, which holds the gate filter at M2. The bias rail comes up at once.
C228: optional Y-cap to BUS- (DNP), fit only with its twin on the other bank, both equal.

Heater supply 12.6 V DC 3.5 A (floating; referenced by the tube string midpoints)



U261 is clocked from W3 itself: U262 compares the winding polarity, C270/C271 + D266/D267 turn both edges into 190 kHz pulses (2 x fs) on RT/CLK, so the buck is locked to the LLC without crossing the isolation.
Check the RT/CLK pulse requirements (amplitude, width) in the TPS54560B data sheet. Heater string midpoints stay grounded at the tubes (amp sheet).

T201



ETD54 N97, see SMPS.md

T201 (SMPS.md): W1 primary, W2 aux, S1 shield and W3 heater in chamber B; W4/W5 (+S2) in chamber A on M1, W6/W7 (+S3) in chamber C on M2.
All secondaries triple-insulated wire. Core strapped to PE.
W5/W7 feed the existing bias rectifiers D5/D6 (UF4007) on the amp sheet.

Winding data: SMPS.md, T201
Draft — values are start values, see SMPS.md

Sheet: /SMPS outputs/
File: smps_sec.kicad_sch

Title: PPP power amplifier 4x KT88 — SMPS outputs (mono)

Size: A2	Date: 2026-10-20	Rev: 1
KiCad E.D.A. 9.0.8		Id: 4/4